

New Zealand Journal of Geology and Geophysics

ISSN: 0028-8306 (Print) 1175-8791 (Online) Journal homepage: www.tandfonline.com/journals/tnzg20

Paleomagnetic results from the Beardmore Glacier Region, Antarctica

J. C. Briden & R. L. Oliver

To cite this article: J. C. Briden & R. L. Oliver (1963) Paleomagnetic results from the Beardmore Glacier Region, Antarctica, New Zealand Journal of Geology and Geophysics, 6:3, 388-394, DOI: 10.1080/00288306.1963.10422070

To link to this article: https://doi.org/10.1080/00288306.1963.10422070

Published online: 16 Jan 2012.

Submit your article to this journal

Article views: 145

View related articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tnzg20

PALEOMAGNETIC RESULTS FROM THE BEARDMORE GLACIER REGION, ANTARCTICA

J. C. BRIDEN

Department of Geophysics, Institute of Advanced Studies, Australian National University, Canberra, Australia

and

R. L. OLIVER

Department of Geology, University of Adelaide, Adelaide, South Australia

(Received for publication, 3 October 1962; as amended, 15 November 1962)

ABSTRACT

A paleomagnetic study has been made of thirteen oriented samples of Jurassic dolerite from sills intruding sedimentary rocks of the Beacon Group near the Beardmore Glacier (84°S, 165°E). Laboratory tests indicate a stable direction of magnetisation with a declination of 244° and an inclination of −75° (steeply upwards), from which the paleomagnetic pole relative to Antarctica for the Jurassic is calculated on the geocentric dipole model to have been at 59°S, 139°W: this is in agreement with results from dolerites of similar age from two other areas in Antarctica. Such demonstrations of uniformity are powerful evidence for the correctness of the geocentric dipole model for the ancient Earth's field, and for the reliability of paleomagnetic results.

INTRODUCTION

Throughout much of Victoria Land and the Ross Sea region generally, a basement complex of metasedimentary and intrusive rocks is overlain unconformably by flat-lying sediments of the Beacon Group (see, for example, Harrington, 1958). Both the basement and the cover, particularly the cover, are cut by extensive dykes and sills known as the Ferrar Dolerites. These have been considered to be of Jurassic age (Harrington, 1958). They can be correlated with some confidence through Victoria Land to the Beardmore Glacier region, and although their correlation across the Pole to the Theron Mountains region is less certain, it is nevertheless considered probable that these dolerites all belong to the same intrusive phase. Recent potassium–argon dating of a dolerite at Lake Vashka in the Victoria Valley at 162 million years (Evernden and Richards, 1962), and elsewhere by McDougall (1963) at from 147 to 163 million years, may be taken as confirmation of the Jurassic age of the dolerites as a whole.

The dolerites have previously been the subject of paleomagnetic studies chiefly in two widely separated regions. Blundell and Stephenson (1959) obtained results from eight oriented samples from the Theron Mountains

region (79–82°S, 25°W). Most observations, however, have been obtained from the dry valleys of the Ferrar Glacier – Wright Valley – Victoria Valley region (78°S, 161°E), and are based on over a hundred samples. Results of the work in the Ferrar Glacier area were reported by Turnbull (1959). Bull and Irving (1960) studied the dolerites in the nearby Wright Valley, a study which has since been extended to the adjacent Victoria Valley (Bull, Irving, and Willis, 1962).

In the last mentioned paper the mean direction of magnetisation measured in five oriented samples of dolerite from a third distinct region (north of the Beardmore Glacier) was also reported. It was stated that the results were in broad agreement with the other regions that had been studied in more detail, but no detailed analysis was attempted because the number of samples collected was small. The five samples were collected by Mr R. I. Walcott on the New Zealand Geological Survey Expedition 1959–60. Subsequently one of us (R.L.O.) collected eight oriented samples from a dolerite sill at Plunkett Point (85° 3½'S, 166°E), near the Beardmore Glacier, while he was a guest of the United States Antarctic Research Programme 1961–62. The directions of magnetisation and some other magnetic properties of these rocks have now been measured, and the results from the two collections have been combined to provide a unified result for the Beardmore Glacier region. The data are now sufficient to make interesting comparisons with the results of the studies mentioned previously, which were in areas respectively 1,000 miles and 450 miles from the Beardmore Glacier, and in opposite directions. Such comparisons, which test the consistency of paleomagnetic results from rocks of comparable age in the same land mass, are of basic importance in assessing the value of those results. At the time of the earlier studies the age of the dolerites was estimated on uncertain grounds, but now that radioactive age determinations are being made on these same rocks the paleomagnetic work will be related to a good time base, making the tests of consistency more valuable.

The samples from the more northerly locality in the Beardmore Glacier region, at 83° 22'S, 164° 10'E, were taken from two sills. Two sites were near the middle of a thick sill intruded low in the Beacon Group, and three were spread through another 300 ft sill intruded higher in the Group. At Plunkett Point the collection was from four sites spread through the bottom 400 ft of a thick sill also intruding rocks of the Beacon Group. The top of this sill has been eroded.

METHOD

Up to three discs were cut from each sample, and the intensity and direction of magnetisation in each disc (26 in all) were measured by an astatic magnetometer. Partial demagnetisation experiments in alternating magnetic fields, on a trial set of Beardmore samples, using the methods and apparatus described by Irving, Stott, and Ward (1961), suggested that the magnetic properties of these dolerites were similar to those described by Bull and Irving (1960) and by Bull, Irving, and Willis (1962) in the dolerites of the Wright and Victoria Valleys. The experiments suggested

that in many specimens the observed directions were made up of two components: (1) a stable component, and (2) a much smaller component parallel to the present geomagnetic field, which could be removed by partial demagnetisation in fields of 150 oersteds (peak). The directions of magnetisation in all specimens were therefore remeasured after treatment in a field of that magnitude.

RESULTS

The results, which are shown in Fig. 1, were analysed according to the method of Fisher (1953). First the mean directions at each site were obtained; these were then combined by giving unit weight to each site, to calculate a mean direction (D_m, I_m) for the dolerites. This was done both for the initial measurements and for the directions obtained after partial demagnetisation. There was found to be a small change in the mean direction, predominantly away from the present field, suggesting that a magnetically soft component parallel to the present geomagnetic field may have been removed by the partial demagnetisation. Statistical analysis showed, however, that this change in the mean direction was not significant at the 95% probability level. Results are summarised in Table 1.

Demagnetisation in alternating fields higher than 150 oersteds produced no significant change in the directions of magnetisation until fields in excess

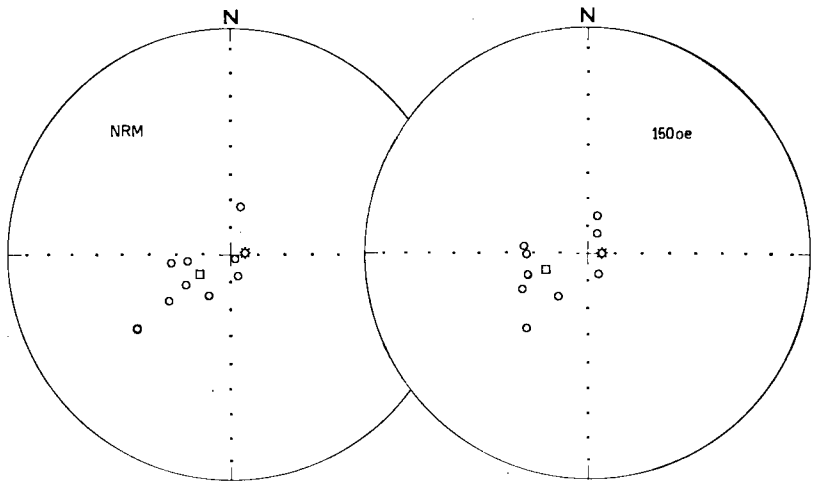


FIG. 1—Mean directions of magnetisation at the 9 sampling sites in dolerites from the Beardmore Glacier region. Site directions denoted by open circles ○; the overall mean is denoted by an open square □. Present Earth's field direction denoted by a star ★. All inclinations are negative (upwards). Equal area projection.

Left: initial directions. *Right:* directions after partial demagnetisation in 150 oersteds (peak).

TABLE 1—Paleomagnetic Results from Dolerites in Antarctica

Study	No. of Samples	No. of Sites	D_m	I_m	k	a_{95}	Pole °W	Pole °S	$d\Psi$	$d\chi$
1. Theron Mountains Region (Initial)	8	8	64	−68	.....	12	136	54	16	16
2. Ferrar Glacier (Initial)	57	5	255	−76	52	3	142	58	5	5
3. Wright and Victoria Valleys (150 oe)	83	46	250	−68	63	3	140	45	4	5
4. Beardmore Glacier (Initial)	13	9	228	−75	18	11	.....	.....	.....	.....
4. Beardmore Glacier (150 oe)	13	9	244	−75	18	11	139	59	18	20

D_m denotes declination (degrees E of North); I_m, inclination (negative upwards from horizontal); k is an estimate of precision; a_{95} is the semi-angle of the cone of confidence at the 95% probability level. $d\Psi$, $d\chi$ are the 95% probability errors in the estimated paleomagnetic pole, measured respectively along and at right angles to the ancient meridian through the site. Sources: (1) Blundell and Stephenson (1959); (2) Turnbull (1959); (3) Bull, Irving and Willis (1962); (4) the present study.

of 600 oersteds were applied, when scattering of directions was observed. This is strong evidence for the stability of the magnetisation as retained after treatment in 150 oersteds.

Measurements were made of the coercivity of maximum remanence (the back field required to nullify the maximum isothermal remanent magnetisation induced in the specimens by applying a direct field—Roquet, 1947; Creer, Irving, and Nairn, 1959). The values ranged between 250 and 600 oersteds. The susceptibilities (σ) (the ratio of the moment induced by an applied direct field to the magnitude of that field) were measured in 0·54 oersteds, which should give a close approximation to the initial susceptibility. The values in the nine samples measured ranged from 30×10^{-6} to 950×10^{-6} e.m.u./cm^3, the mean being 330×10^{-6} e.m.u./cm^3. The intensity (M_n) of the natural remanence ranged from 8×10^{-6} to 530×10^{-6} e.m.u./cm^3 with a mean of 340×10^{-6} e.m.u./cm^3. The ratio M_n/σ ranged between 0·23 and 1·8, and its average was close to unity. These results are consistent with the natural remanence being of stable type. The intensity (M_n), susceptibility (σ), and the ratio M_n/σ, of samples from the contact zone of a sill were considerably less than from the main body of the intrusion. These findings agree with the detailed results presented for other dolerite intrusions by Bull, Irving, and Willis (1962).

DISCUSSION

Results of previous paleomagnetic investigations of Antarctic dolerites are also summarised in Table 1. Because these studies were made on rocks from widely separated areas, the mean directions of magnetisation are not themselves comparable. Paleomagnetic pole positions are calculated in terms of a simple model for the ancient geomagnetic field—corresponding to the field due to a geocentric dipole. Comparisons may now be made on the basis of these poles. Poles calculated from each of the studies listed in Table 1 are plotted in Fig. 2.

It should be noted that the statistical parameters, k, a_{95}, $d\Psi$, and $d\chi$, listed in the Table are not strictly comparable between the different studies. In study No. 2 (see Table 1) each sample was given unit weight. Study No. 1 and the present analysis (No. 4) were limited by small collections, and the procedure of giving each site unit weight was adopted. The higher values of $d\Psi$ and $d\chi$, compared with study No. 2, are a reflection of this more conservative procedure. In study No. 3, k and a_{95} are derived from the two-level analysis of Watson and Irving (1957); this analysis can be used only when numerous well grouped data are available. The value of k given in Table 1 represents the between-site precision : the overall precision derived from this analysis, taking account of both within- and between-site variation, and from which a_{95} is calculated, is 1877.

The results are in broad agreement, but there is one small apparent discrepancy between studies Nos. 2 and 3: it may be noted that the ovals of confidence about the paleomagnetic pole positions calculated from these two studies do not intersect. This difference is due in part to the different

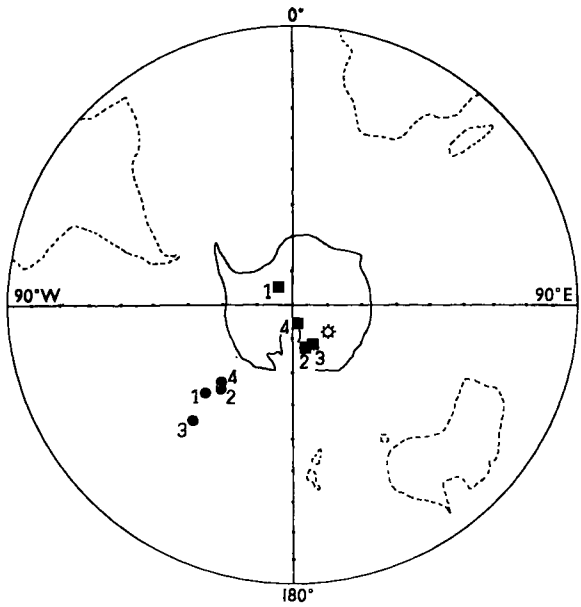


FIG. 2—Southern Hemisphere (equal area polar projection), showing sampling areas, denoted by solid squares ■, and the corresponding pole positions, denoted by solid circles ●. 1 refers to the Theron Mountains Region (Blundell and Stephenson, 1959); 2 and 3 to the Ferrar Dolerite studies of Turnbull (1959), and Bull, Irving, and Willis (1962), respectively; 4 refers to the Beardmore Glacier region. The south geomagnetic pole is denoted by a star ★. The outlines of Southern Hemisphere lands other than Antarctica are dotted.

statistical treatments outlined above, but also to the fact that in one study (No. 2, Turnbull, 1959) no demagnetisation was carried out, and systematically directed unstable components of magnetisation may be present.

CONCLUSION

The evidence presented is, then, consistent with the geocentric dipole hypothesis, and suggests that the sampling areas have belonged to a single crustal unit, at least since the Jurassic. If it is now assumed that the dipole field is directed along the rotational axis of the Earth (geocentric axial dipole hypothesis), then the paleomagnetic poles represent paleogeographic poles. The results from the Jurassic dolerites in Antarctica would then imply a movement of the pole of some 35° relative to the Antarctic continent subsequent to the emplacement of the dolerite intrusions.

Two types of movement might explain this difference between the Jurassic and present pole positions: (a) the continents have remained fixed in the outer shell of the earth while the rotational axis has changed (and hence the poles have shifted), or (b) the rotational axis and poles have remained fixed while Antarctica has moved relative to the poles and to the

other continents. In order to investigate which of these two types of movement has predominated it is necessary to take into account paleomagnetic measurements on rocks of Jurassic age from other continents. There are considerable differences in the Jurassic pole positions obtained from the various southern continents, and in all cases except South America (Creer, 1958) these differ significantly from the present pole position. Illustrations of this have been given by Blundell (1961) and by Bull and Irving (1960). It is clear that polar wandering as defined above is inadequate to explain the discrepancies, but large relative movements of the southern continents could explain the differences, which, as pointed out by Bull and Irving (1960), are much reduced in Du Toit's reconstruction of Gondwanaland.

ACKNOWLEDGMENT

Appreciation is expressed for the opportunity afforded by the United States Antarctic Research Programme for one of us (R.L.O.) to visit Plunkett Point and collect the samples.

REFERENCES

BLUNDELL, D. J. 1961: The Palaeomagnetism of some Igneous Rocks from Antarctica. *Polar Record 10 (67)*: 349–52.

BLUNDELL, D. J.; STEPHENSON, P. J. 1959: Palaeomagnetism of some Dolerite Intrusions from the Theron Mountains and Whichaway Nunataks, Antarctica. *Nature (London) 184*: 1860.

BULL, C.; IRVING E. 1960: The Palaeomagnetism of some Hypabyssal Intrusive Rocks from South Victoria Land, Antarctica. *Geophys. J. 3 (2)*: 211–24.

BULL, C.; IRVING, E.; WILLIS, I. 1962: Further Palaeomagnetic Results from South Victoria Land, Antarctica. *Geophys. J. 6 (3)*: 320–36.

CREER, K. M. 1958: Preliminary Palaeomagnetic Measurements from South America. *Ann. Geophys. 14 (3)*: 373–90.

CREER, K. M.; IRVING, E.; NAIRN, A. E. M. 1959: Paleomagnetism of the Great Whin Sill. *Geophys. J. 2 (4)*: 306–23.

EVERNDEN, J. F.; RICHARDS, J. R. 1962: Potassium-Argon Ages in Eastern Australia. *J. Geol. Soc. Aust. 9 (1)*: 1–49.

FISHER, R. A. 1953: Dispersion on a Sphere. *Proc. Roy. Soc. Lond. A 217*: 295–305.

HARRINGTON, H. J. 1958: Nomenclature of Rock Units in the Ross Sea Region, Antarctica. *Nature (London) 182*: 290.

IRVING, E.; STOTT, P. M.; WARD, M. A. 1961: Demagnetization of Igneous Rocks by Alternating Magnetic Fields. *Phil. Mag. 6 (66)*: 225–41.

MCDOUGALL, I. 1963: Potassium-Argon Age Measurements on Dolerites from Antarctica and South Africa. *J. Geophys. Res. 68 (5)*: 1535–45.

ROQUET, J. 1947: Sur l'Aimantation Rémanente Isotherme du Sesquioxyde de Fer *C.R. Acad. Sci. (Paris) 224*: 1418–20.

TURNBULL, G. 1959: Some Palaeomagnetic Measurements in Antarctica. *Arctic 12 (3)*: 151–7.

WATSON, G. S.; IRVING, E. 1957: Statistical Methods in Rock Magnetism. *Mon. Not. Roy. Astron. Soc. Geophys. Suppl. 7 (6)*: 289–300.